



COLLAPSE

The battering waves carve away coastal cliffs, and loosen big blocks of rock that eventually fall away, undermining the cliffs above. Over time the rock collapses under its own weight, dumping big boulders on the tidal shore below. These absorb most of the force of the waves until they are broken up too, exposing the cliff to further attacks.

▲ ROCKFALL

Undercut by big storm waves, part of this chalk cliff has fallen on to the shore. The mound of rubble will protect the cliff, but not for long.

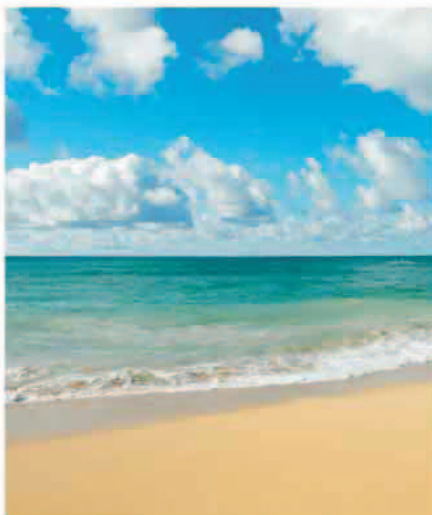


ROLLING AND TUMBLING

As soon as the rock falls onto the shore, the waves start tossing it around. This knocks the corners off, forming rounded boulders, shingle, and sand. The turbulent water sweeps the smaller fragments away, either suspended in the water or bounced and rolled over the seabed, but the large boulders remain where they fell.

SHELTERED BEACHES

On seashores that are sheltered by projecting headlands, the sea is much calmer, with smaller waves. Instead of carving away the shore, the waves build it up by adding loose stones and sand carried along the coast. The relatively calm water cannot shift heavy stones, so these sheltered shores are marked by beaches of fine sand. Where the waves are bigger, they build beaches of larger stones known as shingle. Some of these beaches keep growing, but others are reshaped every year by winter storms.



SHIFTING SAND

As moving water carries rock debris along the shore, it shifts the small, light particles more easily than the bigger, heavier ones, and carries them further from the shore where they fell from the cliff. This tends to sort the debris into different sizes, because the water drops the heaviest pebbles first, followed by smaller shingle, then sand.



Sand

Fine shingle

Shingle

Large pebbles